Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **C++/Python systems**, **multithreaded software**, fault-tolerant workflows, simulation pipelines, and **automated verification** for reliable production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **automation pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production monitoring workflows reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship | *India*
- Reduced centralized-control and quantum-risk exposure by 25% in benchmark simulations through a **novel consensus algorithm**, evaluating protocol behavior and failure trade-offs against established blockchain mechanisms systematically and reproducibly.
- Integrated five lattice-based cryptographic primitives into a **quantum-resistant protocol design**, documenting technical decisions and contributing peer-reviewed research presented at IISc Bengaluru through an IEEE publication with reproducible benchmark evidence.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript, TypeScript
Systems & Algorithms: Data Structures, Algorithms, Operating Systems, CPU Scheduling, Multithreading, Concurrent Programming, Distributed Systems, Object-Oriented Programming, System Design, TCP/IP
Quality & Reliability: Unit Testing, Integration Testing, Regression Testing, Automated Testing, PyTest, Mockito, Debugging, Fault Isolation, Error Handling, Data Validation, Performance Optimization, Observability
Tools & Delivery: Linux, Git, GitHub, Jenkins, CI/CD, Docker, Kubernetes, GitHub Actions, Postman, JIRA, Agile
Backend & Data: FastAPI, Flask, REST APIs, Microservices, PostgreSQL, MySQL, AWS RDS, AWS S3, Cloud Storage, JSON

Projects

Real-Time Instrument Command Runtime - Design Concept
- Designed a **real-time command runtime** concept in C++ with deterministic state transitions, concurrent task handling, command validation, telemetry queues, fault responses, and traceable nominal and off-nominal execution paths.

Flight Software Verification Harness - Design Concept
- Designed a **flight-software verification harness** concept with simulator-driven scenarios, unit and regression suites, repeatable fixtures, fault injection, expected-state assertions, structured test evidence, and CI release gates for reliable delivery.

CPU-Scheduling Simulator | Live
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios.